Syllabus

COSC 1210, Computer Science I, Fall 2026 | Augustana U.

Course Info

- Where: FSC 372 | When: MWF 2:00-3:10 pm
- Canvas: https://augie.instructure.com/courses/15635
- AI use: permissive (details below)

Instructor

- Dr. Marcus Birkenkrahe
- Office/student hours: in room FSC 386 - book via [Navigate]
- Tue 1:00-3:00 pm, Thu 9:00 am-12:00 pm
- Phone: 605.274.4285 | Email: marcus.birkenkrahe@augie.edu
- Bio: birkenkrahe.github.io | linkedin.com/in/birkenkrahe

Course Description

An introduction to computer science, which includes topics such as software engineering, computer architecture, and programming languages. Emphasis on learning the styles, techniques, and methodologies necessary to design and develop readable and efficient programs.

Course AI Classification

AI Use in this Coursework: Permissive.

Students may use AI tools freely throughout this course, including on assessments, unless otherwise specified, provided use is disclosed and outputs are critically evaluated. AI tools are not permitted during exams unless explicitly stated on the exam itself.

Course Outcomes

Students who complete this course will be able to:

- Write, run, and debug Python programs interactively and as scripts
- Represent and manipulate data using variables, expressions, and statements
- Control program flow with conditional execution
- Design functions with parameters and return values to organize code
- Implement iterative solutions using while and for loops, including accumulation patterns
- Process text using string indexing, slicing, and built-in methods
- Store, traverse, and manipulate collections using lists and dictionaries
- Read from and write to text files
- Decompose problems into smaller, well-defined steps
- Read, trace, and explain the behavior of existing code
- Develop small, complete programs independently, applying multiple concepts together

University Policies and Resources

[Use this link for](): University-wide academic policies (Accessibility & Accommodations, Canvas, Honor Code, Illness Prevention & Protection, Student Well-being/Mental Health, Title IX Statement, and other Campus Resources including Help Desk, Registrar's Office, & Writing Center) are covered in a separate university-wide policy document.

Course Policies and Expectations

Optional policies are noted where applicable.

Attendance/Punctuality

This is a face-to-face course and you need to attend class to be successful in this topic. Many of our assignments will be started as in-class activities and will require your presence. If you are absent, make arrangements with a classmate to get class notes and a briefing on discussion topics, watch session recordings if available, and review the online material. Classes will start at the scheduled time and you should arrive 2-3 minutes prior to the start of class.

Absences

Let me know via email if you cannot attend a class. Augustana University only recognizes University-related absences and absences due to medical illness (you must provide a doctor's note).

Generative Artificial Intelligence Policy

AI tools predict, they don't think, and they will confidently hand you wrong answers alongside right ones. Learning to verify, correct, and take responsibility for AI-assisted work - propose, test, find where it breaks, revise - is part of what this course teaches, and it's also just good scientific practice. What isn't allowed is using AI as a substitute for that judgment: undisclosed use, or submitting output you haven't checked, defeats the purpose. For more details on "should you use AI to help you code", see: https://github.com/birkenkrahe/org/blob/master/fall24/UsingAItoCode.pdf

Course Modification (inclement weather, illness, etc.) Policy

If significant inclement weather, professor illness, or any special circumstance arises that may require an adjustment to the course schedule (i.e. an alternative assignment, switching to synchronous online, canceling the class, etc.), every attempt will be made to communicate such changes to students by the night before the scheduled class. Communication to students will come via an announcement on the class LMS and/or an email to your AU email address. If you have reason to believe the class may not be able to be offered as scheduled, please keep an eye on these inboxes.

Exam Policy

Exams during the term are based on material discussed in class. Make-up exams will only be given if a written excuse (a physician's note or evidence of a University-excused absence) is submitted to me prior to the exam. Make-up exams may be different than in-class exams and the material may generally be more difficult. During exams, no cell phones, calculators, or other electronic devices may be used. All notes and books must be out of sight, except for materials allowed (as specified during the review session). You may not leave the room during the exam.

Course Materials and Access

Required textbook

- We are going to use the textbook "Think Python" by Allen B. Downey, 3rd edition. It is freely available online: https://allendowney.github.io/ThinkPython/.

Course platforms

- GitHub for lectures, code: https://github.com/birkenkrahe/cosc_1210_fall_26
- Google Chat Space for discussion and sharing: https://tinyurl.com/gspace-1210
- Google Drive for whiteboard photos and papers: https://tinyurl.com/fa26-photos-1210
- DataCamp for additional optional practice: https://www.datacamp.com
- Jupyter notebooks for code along and assignments: https://jupyter.org/try-jupyter/lab/
- YouTube Playlist: https://www.youtube.com/watch?v=9uW6B9LPntY&list=PLRe9Qsi9YpYU

Assignments and Grading Procedures

Grading table

Letter Grade	Percentage Range
A	93-100%
A-	90-92%
B+	87-89%
B	83-86%
B-	80-82%
C+	77-79%
C	73-76%
C-	70-72%
D+	67-69%
D	63-66%
D-	60-62%
F	Below 60%

Grade breakdown

What	When	How much
Programming assignments	weekly	40%
Quizzes (online)	weekly	25%
Participation	always	10%

What	When	How much
Midterm exam (50 min)	TBD	15%
Final exam (optional)	TBD	10%

- **Programming assignments:** at home, individual, AI allowed unless specified - use must be disclosed, outputs critically evaluated and tested.
- **Quizzes:** multiple choice, not timed, at home, individual
- **Midterm exam:** 50 min, individual, based on quizzes (with mock exam)
- **Final exam:** 30 min, optional, individual conversation

Course Schedule

One quiz and one programming assignment per week (see Grading). This schedule may be subject to changes depending on class progress.

Date	Topic	Notes
Wed (8/26)	Overview	Review the AI policy, clarify questions
Fri (8/28)	Colab and Python	
Mon (8/31)	Ch 1. Programming 1/2	
Wed (9/2)	Ch 1. Programming 2/2	Last day for online reg./adding w/o signature
Fri (9/4)	Ch 2. Variables/Statements 1/2	
Mon (9/7)	NO CLASS - Labor Day	
Wed (9/9)	Ch 2. Variables/Statements 2/2	
Fri (9/11)	Ch 3. Functions 1/2	Independent Study/Internship paperwork due
Mon (9/14)	Ch 3. Functions 2/2	
Wed (9/16)	Ch 4. Functions/Interfaces 1/2	Last day to drop without signatures
Fri (9/18)	Ch 4. Functions/Interfaces 2/2	Extended chapel service (until 11:15a)
Mon (9/21)	Ch 5. Conditionals/Recursion 1/3	
Wed (9/23)	Ch 5. Conditionals/Recursion 2/3	
Fri (9/25)	Ch 5. Conditionals/Recursion 3/3	
Mon (9/28)	Ch 6. Return values 1/2	
Wed (9/30)	Ch 6. Return values 2/2	Graduation Application Deadline (10/1)
Fri (10/2)	Ch 7. Iteration and search 1/3	
Mon (10/5)	Ch 7. Iteration and search 2/3	
Wed (10/7)	Ch 7. Iteration and search 3/3	
Fri (10/9)	MIDTERM	
Mon (10/12)	NO CLASS - Native American Day	

Date	Topic	Notes
Wed (10/14)	NO CLASS - Fall Break	
Fri (10/16)	Ch 8. Strings/Regex 1/2	
Mon (10/19)	Ch 8. Strings/Regex 2/2	
Wed (10/21)	Ch 9. Lists 1/2	
Fri (10/23)	Ch 9. Lists 2/2	
Mon (10/26)	Ch 10. Dictionaries 1/2	
Wed (10/28)	Ch 10. Dictionaries 2/2	Last day to drop with "W" or file S/U
Fri (10/30)	Ch 11. Tuples 1/2	
Mon (11/2)	Ch 11. Tuples 2/2	
Wed (11/4)	Ch 12. Text analysis/generation 1/3	
Fri (11/6)	Ch 12. Text analysis/generation 2/3	
Mon (11/9)	Ch 12. Text analysis/generation 3/3	
Wed (11/11)	NO CLASS - Veterans Day	
Fri (11/13)	Ch 13. Files and databases 1/2	
Mon (11/16)	Ch 13. Files and databases 2/2	Interim 2027 & Spring 2027 registration (11/16-20)
Wed (11/18)	Ch 14. Classes and functions 1/2	
Fri (11/20)	Ch 14. Classes and functions 2/2	
Mon (11/23)	Ch 15. Classes and methods 1/2	
Wed (11/25)	NO CLASS - Thanksgiving Break	
Fri (11/27)	NO CLASS - Thanksgiving Break	
Mon (11/30)	Ch 15. Classes and methods 2/2	
Wed (12/2)	Ch 16. Classes and objects 1/2	
Fri (12/4)	Ch 16. Classes and objects 2/2	Last day of classes
(12/7-12/11)	FINAL EXAM	Date/time/location TBD
		Final grades due Wed (12/16)

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